Examiner only

7. (a) State Newton's **three** laws of motion **and** explain how they each apply to the motion of skydivers from when they first jump from a plane until they first reach terminal speed. [6 QER]

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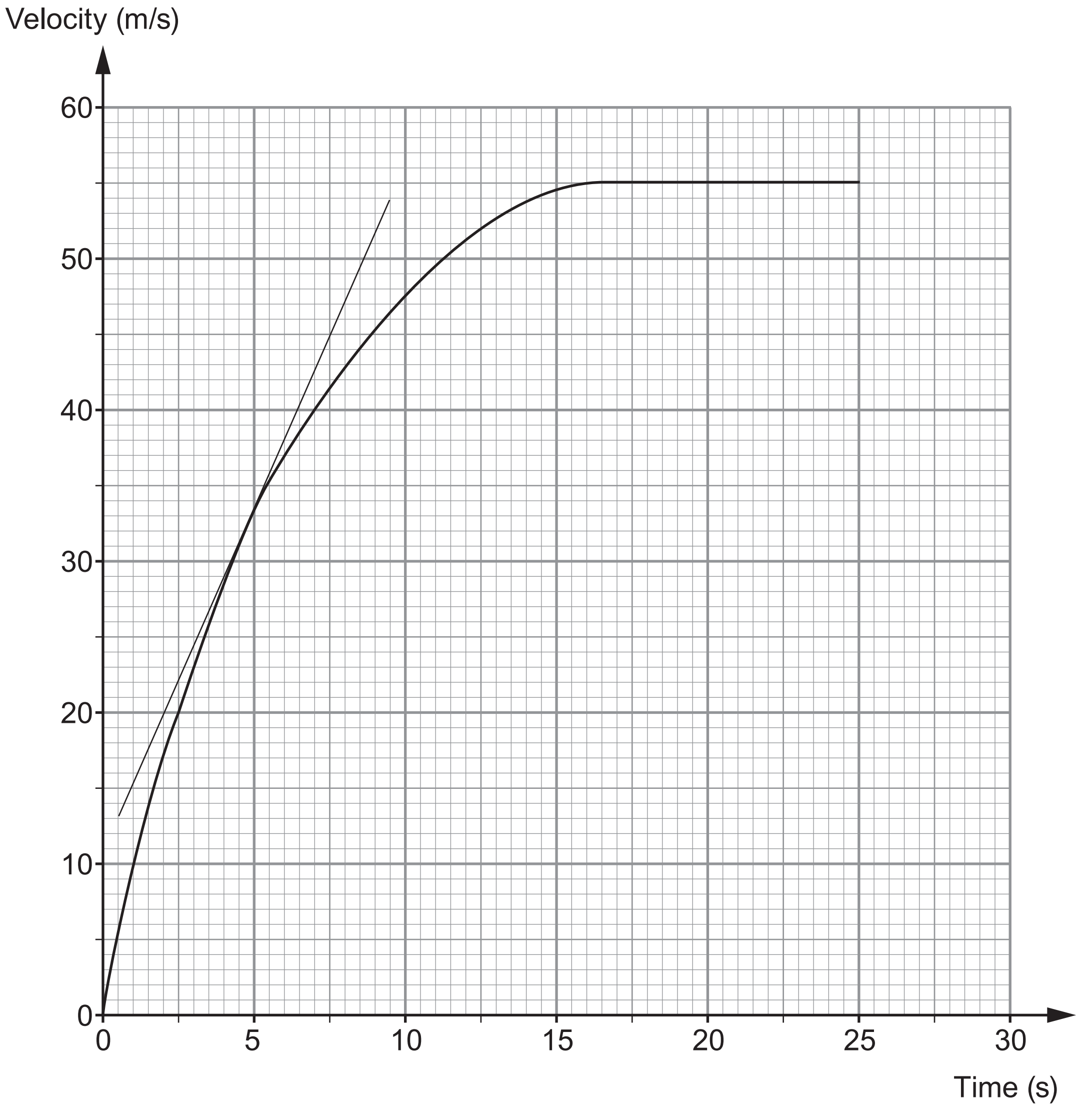
(b) The velocity-time graph below shows the motion of a skydiver.
A tangent has been drawn at 5 seconds.



Examiner only

(i) Acceleration can be calculated by measuring the gradient of a velocity-time graph. Calculate the acceleration of the skydiver at **5 s** by using the tangent shown. [3]

Acceleration = ... m/s^2

(ii) Describe how the acceleration changes over the 25 s shown. [2]

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(iii) Use the graph and an equation from page 2 to estimate the distance travelled by the skydiver in the **first 5 s**. [3]

Distance travelled = ... m

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